

COSE471: Data Science

Course Introduction

Department of Computer Science and Engineering
Korea University
Susik Yoon

Friday, March 6, 2026

Course Schedule

Week	Day	Content	HW/Project	Note
1	Mar 06	Introduction		
2	Mar 13	Data		
3	Mar 20	Pattern mining		
4	Mar 27	Tutorial 1	● HW1	<i>Latex , Preprocessing/Pattern mining</i>
5	Apr 03	Clustering	●	
6	Apr 10	Outlier Detection & Invited Talk	●	<i>An invited talk from Meta</i>
7	Apr 17	Tutorial 2	● HW2	<i>Clustering, Outlier detection</i>
8	Apr 24	Classification	●	
9	May 01	Classification	●	
10	May 08	Tutorial 3	●	<i>Classification, Feature engineering</i>
11	May 15	Mid-term exam		<i>Offline, Closed-book, No electronic devices</i>
12	May 22	Review & Project Notice	● Project	
13	May 29	Project discussion	●	
14	Jun 05	Project discussion	●	
15	Jun 12	Final presentation	●	
16	Jun 19	Wrap-up and Final report	●	

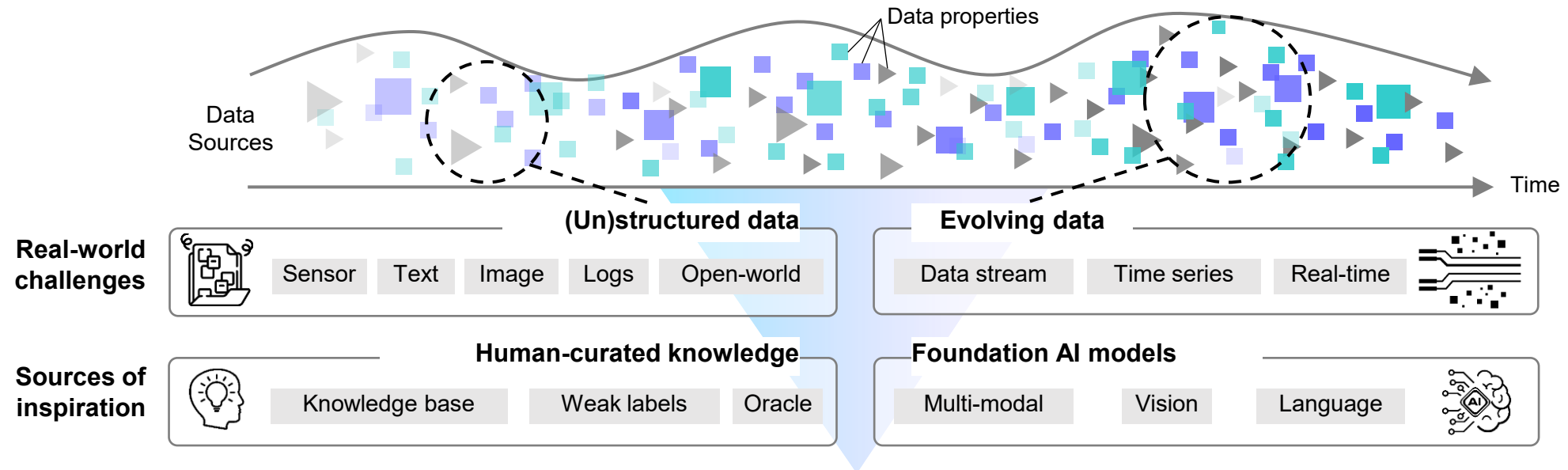
Instructor

□ Susik Yoon (susik@korea.ac.kr)

- Assistant professor, Computer Science and Engineering, Korea University
- Office: Sci&Eng Library 616C / Office hour: by appointment

□ Data and Adaptive Intelligence Systems (DAIS) Lab (<http://dais.korea.ac.kr>)

:= Evolving/unstructured data + DS methodologies + AI foundation models + Human knowledge

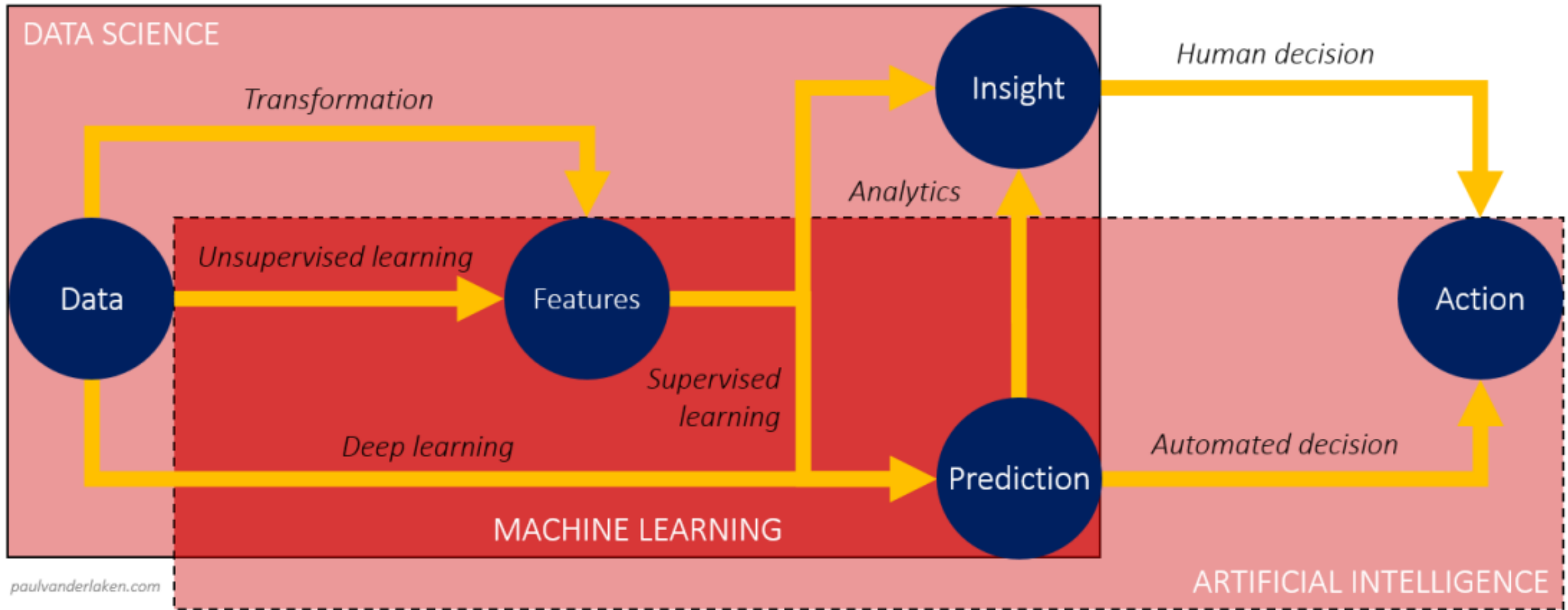


Teaching Assistants

☐ **Awesome TAs who will help you with ...**

- Tutorials
 - Tutorial 1: Hyunwoong, Yerin, Seungwon
 - Tutorial 2: Yoonhwan, Juhyeong, Minyeol
 - Tutorial 3: Subin, Yebin, Myeongchun
- Homework: Woojun, Jeehyo
- Project: Huijeong

Data Science

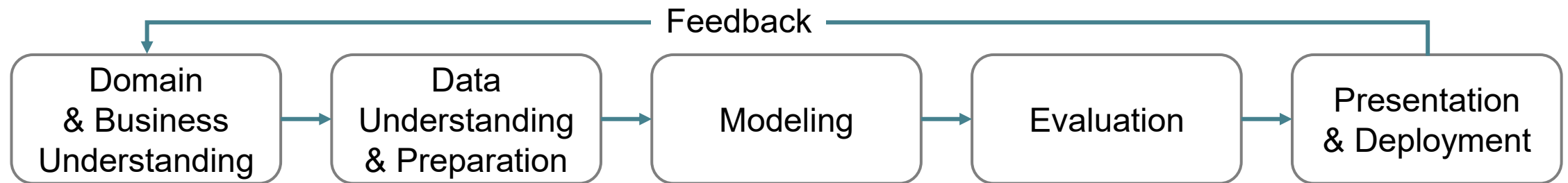


Data Science

□ Definition

- “An interdisciplinary academic field that uses statistics, scientific computing, scientific methods, processes, algorithms and systems to *extract or extrapolate knowledge and insights from potentially noisy, structured, or unstructured data.*”
- The process of *discovering interesting patterns and knowledge* from large amounts of data
 - a.k.a. data mining, knowledge mining from data, knowledge discovery from data, pattern discovery, knowledge extraction, data analytics, and information harvesting

□ Typical procedure

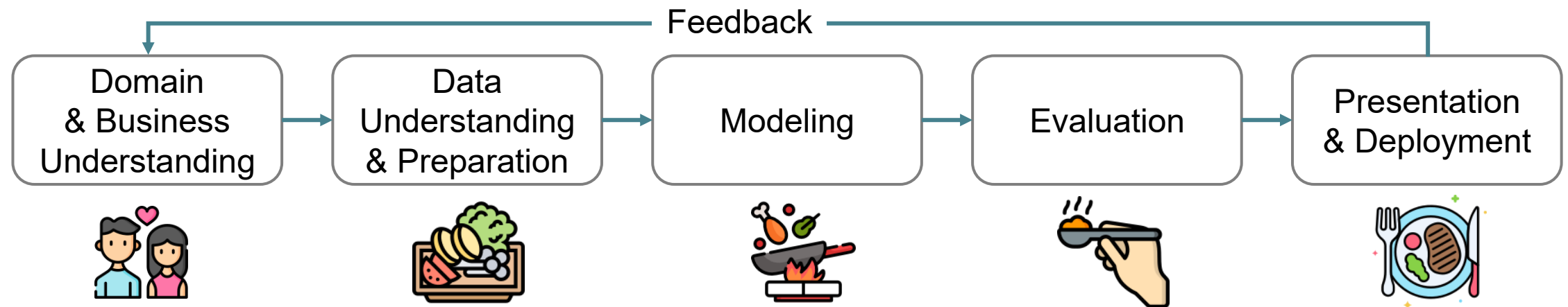


Data Science (like Cooking?)

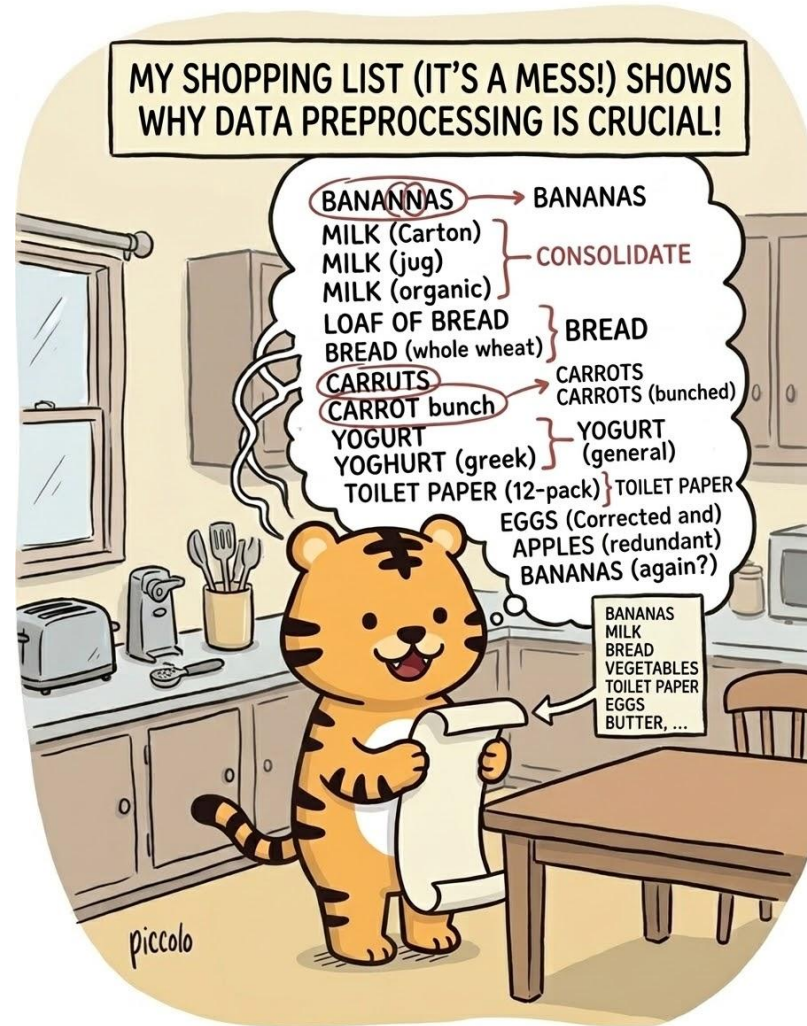
❑ Definition (*of cooking*)

- The process of **discovering interesting patterns and knowledge** from large amounts of data
- (*i.e., the process of transforming raw ingredients into a flavorful and well-presented dish*)

❑ Typical procedure (*of cooking*)



Data Understanding and Preprocessing



Pattern Mining



Clustering



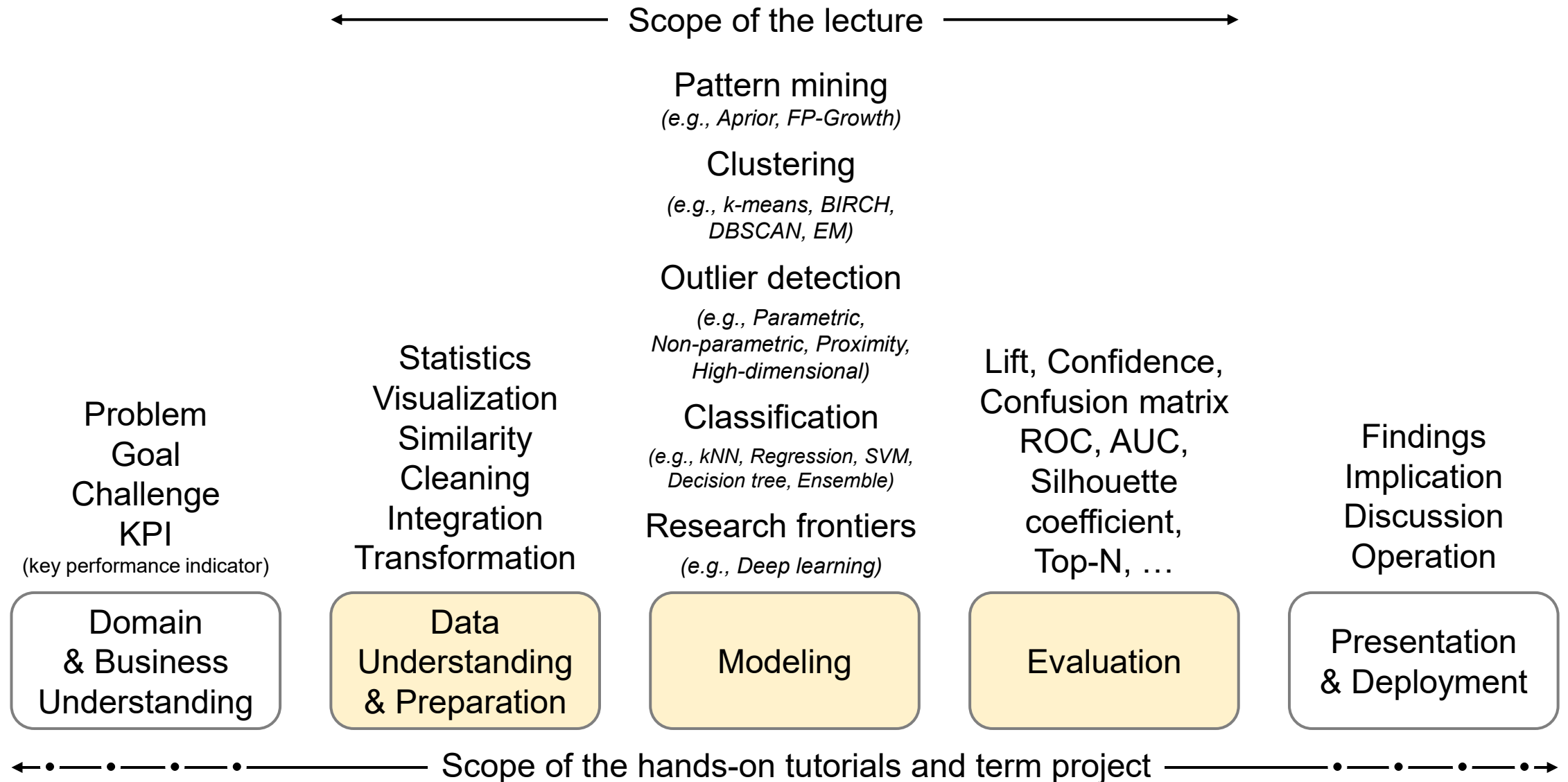
Outlier Detection



Classification



Course Coverage



Course Logistics

❑ Lecture notes will be self-contained in most cases

- Reference: Data Mining: Concepts and Techniques 4th edition by Jiawei Han et al.

❑ Grading

- Attendance (10%)
 - You may miss up to 5 classes; each absence gets you a penalty of -2%
 - Missing 1/3 of the classes (i.e., more than 5 absences) will result in failing the course, following the school policy
- Homework assignments (10%)
- A SINGLE offline exam (50%) on May 15th
- Term project (30%)
 - Team of 5 members
 - Presentation and final report

Discussion for Attendance Check

- ❑ Please share your thoughts on the following topics
 - The reason you chose this course
 - One problem in your daily life you may address with the concepts learned in this class